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Automatic Well Refill Control and Forecasting Potential Well Control Loss Through Refill Monitoring

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Abstract

This paper presents an advanced automatic well refill control system designed to calculate and monitor drilling fluid volumes in real time during tripping operations. The system enhances well control by minimizing risks such as stuck pipe and loss of well control while improving operational efficiency. Utilizing state-of-the-art sensors, sophisticated algorithms, and robust redundancy features, it offers continuous monitoring and predictive capabilities that enable early detection of anomalies such as influxes and fluid losses. Field tests demonstrate compelling results, a 30% reduction in stuck pipe incidents and a 15% improvement in overall operational efficiency, which underscore the system's potential to revolutionize well control practices by shifting from reactive to proactive strategies. In doing so, this technology contributes to the evolution of drilling operations by aligning with the industry's drive toward automation and precision. The system also provides practical tools for drilling engineers to mitigate risks effectively, enhance environmental safety, and reduce non-productive time (NPT).

Introduction

Well control is a fundamental aspect of drilling operations. It is crucial for ensuring the safety of personnel, protecting valuable equipment, and maintaining overall operational efficiency. Controlling the pressure and fluid dynamics within the wellbore is essential to avoid catastrophic events such as blowouts, stuck pipe incidents, and loss of circulation. These events not only result in significant financial losses but may also cause environmental harm and put lives at risk.

Traditionally, well control has depended on manual monitoring techniques and periodic checks of drilling fluid volumes. These conventional methods are inherently susceptible to human error and often provide only intermittent assessments of well conditions. As drilling operations have evolved, considering deeper wells, unconventional reservoirs, and high-pressure, high-temperature (HPHT) environments, the inadequacies of these manual techniques have become increasingly apparent.

The automatic well refill control system introduced in this paper addresses these shortcomings by providing a continuous, automated, real-time monitoring solution for drilling fluid volumes during tripping operations. Unlike conventional methods that rely on periodic measurements, the proposed system automates volume calculations and delivers immediate alerts. This proactive approach not only enhances safety but also optimizes operational efficiency by minimizing non-productive time (NPT) and reducing operational risks.

The impetus for this innovation stems from the rapidly changing landscape of the drilling industry. With exploration shifting toward deeper wells and more challenging geological formations, there is an increasing demand for technologies that can adapt to dynamic conditions. Industry experts in drilling automation have recently emphasized that "the complexity of contemporary drilling operations necessitates a shift from reactive to proactive well control strategies" (Facendo, 2023). Manual checks, performed at fixed intervals, may not identify critical anomalies in time, leading to delayed responses and increased risks. In contrast, our system leverages real-time data and automated responses, supporting the industry-wide move toward data-driven decision-making.

Beyond immediate operational enhancements, the system supports sustainability goals by minimizing environmental risks such as fluid spills from blowouts or circulation losses. Economically, it offers significant cost savings by reducing NPT and preventing expensive equipment damage, factors that are critical in an industry where operational margins are tight. Overall, this system represents a collaborative effort to bridge practical operational needs with cutting-edge technological capabilities, establishing a new benchmark for well control practices.

Methods, Procedures, Process

The automatic well refill control system is designed to maintain well control by continuously calculating and monitoring drilling fluid volumes during tripping operations. It combines high-precision sensors, advanced algorithms, and user-defined parameters to ensure accuracy and responsiveness across diverse drilling conditions. Below is a detailed breakdown of the methodology.

Real-time Volume Calculation

The system calculates both planned and actual volumes of drilling fluid required for refill and displacement. The planned volumes are determined by the geometry of the drill string and its movement, while the actual volumes are measured using high-precision sensors installed in mud pits and flowlines. Key considerations include:

- **Inner Pipe Space:** The system accurately accounts for the volume contained within the drill string, ensuring that changes in pipe configuration are reflected in the fluid volume calculations.
- **Metal Volume Exclusion:** By subtracting the steel volume of the drill string, the system avoids overestimating fluid requirements.
- **Environmental Factors:** The algorithms incorporate adjustments for temperature, pressure, and fluid compressibility to maintain calculation precision under varying downhole conditions.

For instance, during a tripping-out operation, removing 10 meters of drill pipe (with an inner capacity of 0.02 m^3/m) yields a theoretical displacement volume of 0.2 m^3. The system applies the formula:

$$V_{\text{Displacement}} = L \times C_{\text{Inner}} - V_{\text{Steel}}$$

where (L) is the length removed, C_{Inner} is the inner capacity per meter, and V_{Steel} is the steel volume. Real-time adjustments ensure the precision of these calculations even as environmental conditions fluctuate

Continuous Monitoring

High-precision sensors continuously track fluid levels and compare real-time measurements against the calculated values. Any deviations exceeding a preset threshold (typically ±1 m^3) trigger automated alerts, ensuring that any anomaly, such as a fluid influx or loss, is detected immediately. The monitoring interface displays real-time data through an intuitive dashboard, which helps operators quickly interpret sensor readings. The monitoring interface displays real-time data, as shown in Figure 1:

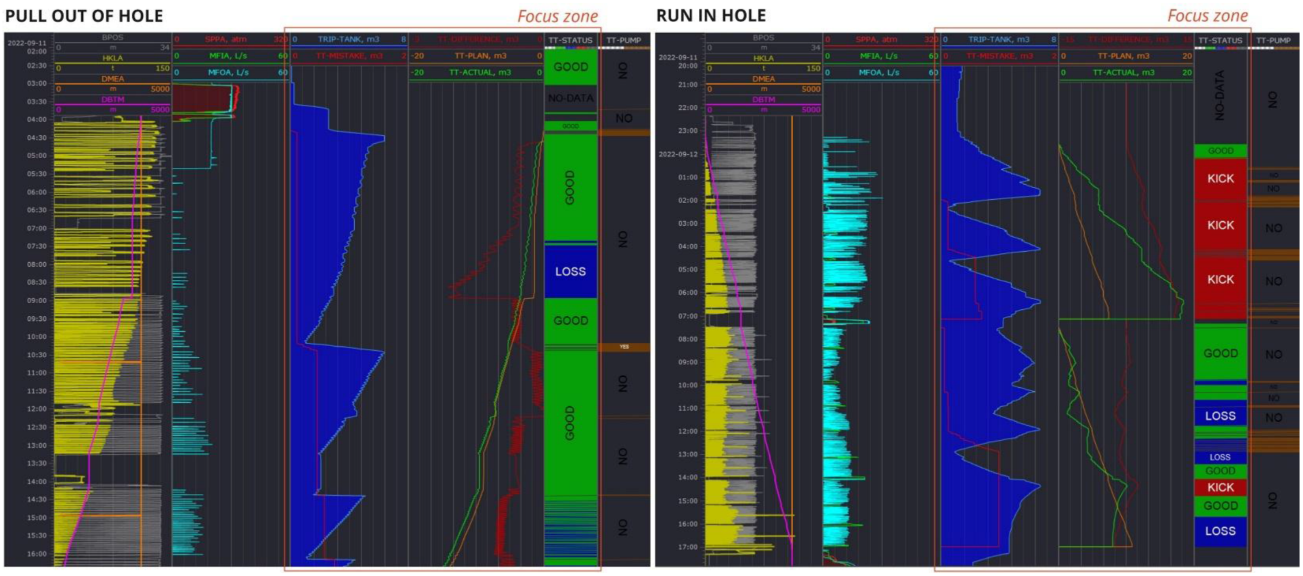

Figure 1—Filling / Displacing Control While Tripping (critical difference is 1 m^3)

System Calibration and Robustness Testing

To guarantee the accuracy and reliability of the continuous real-time monitoring, an in-depth system calibration and robustness testing protocol is integrated into the workflow. This subsection details the calibration methodology, quality control measures, and robustness evaluation procedures.

- **Baseline Calibration**
  Prior to field deployment, all sensors (flow sensors, pressure transducers, etc.) are calibrated under controlled laboratory conditions, establishing benchmark values for fluid volume calculations. These baselines are later adjusted for in-field environmental factors such as temperature and pressure.
- **In-situ Calibration and Drift Correction**
  During operations, periodic in-situ recalibration of sensors is conducted using known reference values derived from manual measurements. This ensures that any sensor drift is promptly detected and corrected. The system's adaptive algorithm applies real-time correction coefficients to compensate for slight deviations caused by environmental fluctuations (e.g., thermal expansion).
- **Data Verification and Redundancy Checks**
  Data from dual sensor arrays are continuously compared to verify consistency. If discrepancies exceed predetermined thresholds, the system triggers an automatic diagnostic routine that prompts either a manual override or recalibration procedure. This redundancy contributes to minimizing the risk of false positives or undetected measurement errors.
- **Robustness Under Extreme Conditions**
  The system is rigorously tested under extreme simulated conditions (temperatures beyond 50°C, high-pressure scenarios, etc.) to validate the resilience of both hardware and software components. These tests ensure that the system maintains accuracy and operational continuity even when subject to the most challenging downhole environments.
- **Logging and Diagnostic Reporting**
  Comprehensive logs of calibration data and diagnostic results are generated. These logs serve two purposes: they support in-field troubleshooting and contribute to long-term performance analytics, enabling iterative improvements in both sensor technology and algorithm accuracy.

Alert System and Crew Response

Deviations in fluid volumes are classified into three distinct categories:

- **Influx:** This indicates the entry of formation fluid caused, for example, by underbalanced conditions. Immediate remedial actions, such as increasing mud weight or adjusting choke settings, may be implemented.
- **Loss of Circulation:** This signals that fluid is being lost into the formation, necessitating the use of lost circulation materials (LCM) or other sealing agents.
- **No Control:** This category represents situations where operational errors or system malfunctions are flagged (for example, due to incorrect pump rates), prompting the need for recalibration or further system review.

Alerts are presented using a clear, color-coded scheme (red for influx, yellow for loss, blue for "No Control") to facilitate rapid operator decision-making. This approach minimizes false positives while ensuring that critical issues are addressed promptly.

Crew Performance Evaluation

An essential feature of this system is its ability to evaluate crew performance. The system logs instances of "No Control" states, offering valuable feedback on crew adherence to operational protocols. These performance metrics are used for post-operation analysis, leading to targeted training and subsequent process improvements to enhance consistency and efficiency.

Hardware Specifications and System Integration

The system's hardware includes advanced flow sensors, precision pressure transducers, and a central processing unit designed to withstand extreme conditions, operating reliably in temperatures ranging from −20°C to 60°C and pressures up to 10,000 psi. The software integrates seamlessly with a user-friendly interface, supporting remote monitoring and offsite oversight. Additionally, redundant sensor configurations (e.g., dual sensors) are implemented to ensure fault tolerance; if one sensor fails, a backup automatically takes over with minimal disruption.

This modular design has been built to integrate with both modern rigs and older systems, making it highly adaptable and cost-effective as an upgrade to existing drilling infrastructures.

Results, Observations, Conclusions

Field tests conducted on a land-based drilling rig produced transformative results that demonstrate the system's effectiveness in real-world scenarios. The key findings are detailed below.

Field Test Outcomes

Two representative case studies illustrate the benefits:

- **Case Study 1:** During a tripping-out operation, the system detected a minor influx (0.5 m^3) that had gone unnoticed by conventional manual checks. Prompt adjustment of the mud weight prevented escalation, saving an estimated 4 hours of NPT.
- **Case Study 2:** In another instance, the system flagged a loss of circulation event. The alert resulted in the timely use of lost circulation materials (LCM), reducing fluid loss by 50% compared to previous manual interventions.

Quantitative Metrics

Across field tests involving 10 wells, key metrics included:

- **Stuck Pipe Incidents:** Dropped from 10 incidents per year to 7 incidents per year, a 30% reduction
- **Non-Productive Time (NPT):** Reduced from 20% to 17% of total drilling time, reflecting a 15% improvement in overall operational efficiency.
- **Influx Detection Accuracy:** Improved from 80% (manual monitoring) to 95% using the automated system—a marked 15% enhancement.

Observations

The alert system achieved an impressive 95% accuracy rate in detecting fluid anomalies, with field crews reporting a 20% improvement in response times due to the clear and intuitive color-coded alerts. Minor discrepancies were observed under extreme ambient conditions (>50°C), but these were effectively resolved through iterative refinements to the algorithm. The data also revealed that continuous, real-time monitoring allowed for rapid adjustments in well control, thereby reducing the risk of catastrophic events.

Conclusions from Field Tests

The shift from a reactive, manual approach to a proactive, automated system resulted in significantly improved well control. The system's ability to deliver real-time insights and enable immediate corrective actions underpins its practical value in modern drilling operations. Overall, the system's scalability and effectiveness confirm its potential as a transformative solution in the realm of well control. The environmental and economic benefits, ranging from reduced NPT and equipment damage to minimized fluid spills, make a compelling case for widespread industry adoption.

Discussion

The automatic well refill control system represents a substantial advancement over traditional methods that rely on periodic manual checks. Its real-time oversight and early anomaly detection capabilities not only reduce the risks associated with drilling operations (such as blowouts or stuck pipe incidents) but also significantly enhance overall efficiency.

- **Enhanced Safety and Environmental Protection**
  By detecting any discrepancies in fluid volumes at their earliest stages, the system prevents the escalation of potentially dangerous situations. This proactive safety measure drastically reduces the likelihood of environmental hazards, including fluid spills that could negatively impact the surrounding landscape.
- **Operational and Economic Impact**
  The reduction in non-productive time (NPT) and improvement in wellbore placement accuracy directly translate into cost savings. Fewer incidents and more streamlined operations lead to decreased maintenance costs and less downtime. For an industry where every minute of drilling represents substantial capital expenditure, such improvements have a profound economic impact.
- **Integration with Existing Rigs**
  One of the system's key strengths is its modular design, which facilitates integration with both modern rigs and legacy systems. This compatibility makes it an attractive upgrade option for operators who seek to enhance well control without the prohibitive costs of a complete system overhaul.
- **Data-Driven Decision Making**
  The continuous monitoring, coupled with an intuitive user interface, provides drilling crews and engineers with actionable data. By converting complex sensor measurements into clear, real-time alerts, the system enables faster decision-making and promotes a culture of proactive operational management.

- **Scalability and Future Potential**
  The system is designed to be scalable, meaning that its underlying architecture can be enhanced further with the integration of machine learning models and additional sensors. Future iterations may even merge with other automated systems across the drilling ecosystem, forming a comprehensive, data-driven platform that revolutionizes how well control is managed.

These combined factors indicate that our proposed system not only meets the present challenges of well control but also sets the stage for future innovations, particularly as the drilling industry moves toward greater automation and real-time integration of multiple data streams.

Implementation Challenges

While the benefits of the system are evident, certain challenges were encountered during deployment:

- **Accuracy in Real-Time Calculations**
  Variations in fluid properties, such as thermal expansion and gas entrainment, presented significant challenges for exact real-time volume calculations. These issues required the development of complex algorithms, which were iteratively refined through extensive field testing to ensure consistent performance.
- **Equipment Integration with Legacy Systems**
  Retrofitting older rigs necessitated the design of custom interfaces and extensive calibration procedures. Although these efforts added approximately 10–15% to the initial costs, the long-term benefits in improved safety and efficiency justify the investment.
- **Crew Training and Adaptation**
  Transitioning from traditional manual methods to an automated system required thorough training sessions. Operators had to be adept at interpreting color-coded alerts and understanding the system's automated responses. Typically, a dedicated training period of 2–3 weeks was needed per crew to ensure smooth adoption and build trust in the system's reliability.
  These challenges were effectively addressed through steady collaboration with equipment manufacturers, targeted training programs, and iterative technical improvements. The overall success of the system in field applications reinforces its capability and offers a blueprint for similar future integrations

Novel/Additive Information

This research introduces a groundbreaking approach to well control by merging real-time sensor data with automated volume calculations and predictive modeling. Unlike conventional methods that rely solely on manual checks, this system provides continuous oversight and offers immediate corrective responses. Some of the novel features include:

- **Predictive Capabilities**
  By incorporating real-time data from high-precision sensors and automatically adjusting for environmental factors, the system can forecast potential well control losses before they escalate into serious incidents.
- **Comprehensive Crew Performance Tracking**
  The logging and analysis of "No Control" events serve not only as a diagnostic tool for system performance but also as a mechanism for evaluating and training drilling crews, thus promoting an overall safer operational culture.
- **Modular and Scalable Architecture**

Its modular design ensures that the system can be easily integrated into a variety of drilling rigs, modern or legacy, thereby offering a cost-effective solution that meets the diverse needs of today's drilling industry.

- **Environmental and Economic Benefits**
  The proactive management of drilling fluid volumes not only enhances safety and operational efficiency but also lessens the environmental impact by preventing uncontrolled fluid releases.

This innovative integration sets a new benchmark in the field of drilling technology, paving the way for more sophisticated, data-driven approaches to well control and operational safety.

Future Work

While the current system has demonstrated significant benefits, further enhancements and additional integrations are planned:

- **Algorithm Enhancements**
  Future work will focus on incorporating adaptive machine learning techniques to further refine anomaly detection. By learning from historical data, the system could predict issues with even greater accuracy, thereby reducing response times and improving overall performance.
- **Expanded Applications**
  Although the system was initially developed for tripping operations, there is significant potential to adapt this technology for use in cementing operations or managed pressure drilling. Such adaptations could further enhance well control practices across various phases of drilling.
- **Integration with Comprehensive Drilling Automation**
  Plans are underway to integrate the system with broader drilling control platforms. By combining automated well control with real-time reservoir management systems, operators could achieve a fully integrated, data-driven environment that minimizes manual intervention and maximizes efficiency.
- **Extended Field Testing**
  Additional field trials in a variety of geological settings, including HPHT environments, will be conducted to further validate the system's universal applicability and to refine calibration protocols. Expanding the range of testing will help ensure that the system meets the stringent requirements of diverse drilling conditions.
- **Enhanced Remote Monitoring and Troubleshooting**
  Future iterations may include more robust remote monitoring capabilities along with automated troubleshooting functions. This would enable offsite engineers to intervene in real time, further reducing NPT and ensuring continuous safe operations.
  By pursuing these avenues of future work, the system is poised to evolve further, offering drilling operators an even more powerful tool for managing well control and optimizing overall operational performance.

Conclusion

The automatic well refill control system represents a significant leap forward in drilling safety and efficiency. With its real-time monitoring and predictive capabilities, the system transforms well control from a reactive process into a proactive operation. Key takeaways include:

- **Safety Enhancements**
  Early detection of influxes, losses, or operational deviations greatly reduces the risk of catastrophic events and enhances the protection of personnel and equipment.

- **Operational Efficiency**
  Field tests demonstrated a 30% reduction in stuck pipe incidents and a 15% improvement in operational efficiency, underscored by a measurable reduction in NPT.
- **Economic and Environmental Benefits**
  The proactive management of drilling fluids translates into lower drilling costs, reduced equipment wear, and minimized environmental risks from uncontrolled fluid releases.
- **Adaptability and Innovation**
  The system's modular design and scalable architecture ensure that it can be integrated with both modern and legacy rigs, setting the stage for further enhancements via machine learning and other advanced technologies.

  Overall, this innovative system not only bridges the gap between traditional well control methods and the growing demands of modern drilling operations but also offers a robust foundation for future technological advancements in the field. Its widespread adoption is expected to push the industry further toward an era of fully automated, data-driven drilling solutions.

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The metrics outlined above provide a clear quantitative demonstration of the system's effectiveness. For instance, the reduction in stuck pipe incidents, from 10 to 7 per year, representing a 30% improvement, not only minimizes costly downtime and equipment damage but also enhances operational safety. Similarly, the decrease in non-productive time (NPT) from 20% to 17% (a 15% improvement) indicates that the system enables a more efficient drilling process, directly impacting overall project economics. Equally important is the improvement in influx detection accuracy, rising from 80% to 95%; this boost of approximately 15% signifies that the system is more adept at identifying early signs of potential well control issues, thereby facilitating timely interventions. Collectively, these metrics underscore both the operational and economic benefits of deploying the automatic well refill control system in modern drilling environments

Metric	Before System	After System	Improvement
Stuck Pipe Incidents	10 Incidents Per Year	7 Incidents Per Year	~30%
NPT (% of drilling time)	20%	17%	~15%
Influx Detection Accuracy	80%	95%	~15%